Lecture 2: Digital Logic Fundamentals

1. Boolean Algebra and Logic Gates

Boolean Algebra Overview

- Boolean algebra is a branch of algebra where variables take binary values: 0 (False) or 1 (True).
- Operations include:
  - **AND (· or ∧)**: Output is 1 if **both** inputs are 1.
  - **OR (+ or ∨)**: Output is 1 if **at least one** input is 1.
  - **NOT (¬ or ¯)**: Inverts the input.
  - **XOR (⊕)**: Output is 1 if **inputs are different**.
  - **NAND**: Output is the inverse of AND.
  - **NOR**: Output is the inverse of OR.

Truth Tables

A	B	A·B (AND)	A+B (OR)	¬A (NOT)	A⊕B (XOR)	A NAND B	A NOR B
0	0	0	0	1	0	1	1
0	1	0	1	1	1	1	0
1	0	0	1	0	1	1	0
1	1	1	1	0	0	0	0

2. Combinational Logic Circuits

Combinational circuits produce outputs based solely on current inputs.

2.1 Adders

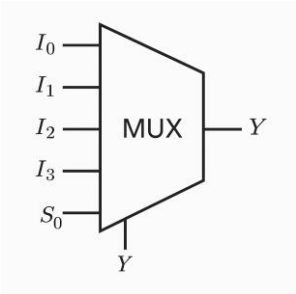
- **Half Adder**: Adds two bits (A, B), outputs Sum and Carry.
  - Sum = A ⊕ B
  - Carry = A · B
- **Full Adder**: Adds three bits (A, B, Cin), outputs Sum and Cout.
  - Sum = A ⊕ B ⊕ Cin
  - Carry = (A · B) + (Cin · (A ⊕ B))

2.2 Decoders

- A decoder converts binary input to a one-hot output.
  - Example: 2-to-4 decoder takes 2 inputs and activates 1 of 4 outputs.
  - Used in memory addressing and control units.

2.3 Multiplexers (MUX)

- A multiplexer selects one input from multiple inputs based on selector bits.
  - Example: 4-to-1 MUX with inputs I0–I3 and selectors S1, S0.
  - Output = I0·¬S1·¬S0 + I1·¬S1·S0 + I2·S1·¬S0 + I3·S1·S0



3. Sequential Logic Circuits

Sequential circuits have memory; output depends on current inputs **and** previous states.

3.1 Flip-Flops

- **SR Flip-Flop**: Set-Reset latch; has undefined state when both S and R = 1.
- **D Flip-Flop**: Data or Delay flip-flop; captures input D on a clock edge.
- **JK Flip-Flop**: Resolves SR undefined state; toggles when J=K=1.
- **T Flip-Flop**: Toggles state on clock if T=1.

3.2 Registers

- A register is a group of flip-flops used to store multi-bit values.
- Common types: Shift registers, Parallel load registers.

3.3 Counters

- Count clock pulses in binary.
- **Asynchronous (Ripple) Counters**: Slower, clock not synchronized.
- **Synchronous Counters**: All flip-flops clocked simultaneously; faster.

4. Basic Building Blocks of a CPU (Overview)

Arithmetic Logic Unit (ALU)

- Performs arithmetic (add, subtract) and logic (AND, OR, NOT) operations.
- Controlled by an **opcode** (operation code) input.

Control Unit

- Directs the flow of data and execution of instructions.
- Decodes instructions and signals other components (e.g., ALU, registers).

Registers in CPU

- **Accumulator**: Holds intermediate results.
- **Program Counter (PC)**: Tracks next instruction.
- **Instruction Register (IR)**: Stores current instruction.

Bus System

- Wires for transferring data:
    - **Data Bus**, **Address Bus**, **Control Bus**

5. Activities: Simple Logic Circuit Analysis

Activity 1: Truth Table for a Given Expression

- Given: $Y = A \cdot B + \neg A \cdot C$
    - Build a truth table and identify output combinations.

Activity 2: Logic Circuit Construction

- Construct a Half Adder using basic gates.
    - Identify where XOR and AND gates are used.

Activity 3: Analyze a 2-to-4 Decoder

- Label inputs and outputs.
- Determine output for input combinations (00, 01, 10, 11).

Activity 4: D Flip-Flop Behavior

- Create a timing diagram showing how output Q changes with clock and D inputs.

Conclusion

- Logic gates form the foundation of digital systems.
- Combinational and sequential circuits are essential for operations in a CPU.
- Understanding these elements provides insight into digital system design and architecture.